Week 10 APP DEV-2

2024 Dec22 ET

1. [Source Q9; MCQ]

Which of the following is true regarding long polling?

(a) Long polling opens multiple connections between the client and the server,and the server continuously sends updates to the client in real-time.

(b) Long polling involves the client repeatedly sending requests at fixed intervalsto the server to check for updates, regardless of whether new data is available.

(c) Long polling allows the client to make a single request to the server, where theserver holds the connection open until new data is available and then sends the response.

(d) Long polling uses WebSockets to maintain a persistent, bidirectionalconnection between the client and the server.

[Source ans: c]

2025 Apr13 ET

2. [Source Q6; MCQ]

Which of the following is a key difference between Web Sockets and Server-Sent Events (SSE)?

(a) SSE allows bidirectional communication, while WebSockets only allow server-to-client communication.

(b) WebSockets maintain a persistent connection, while SSE creates a newconnection for each message.

(c) SSE works over HTTP and only supports server-to-client communication,whereas WebSockets provide full-duplex communication.

(d) WebSockets require an HTTP request for every message sent.

[Source ans: c]

3. [Source Q7; MCQ]

Which HTTP method is typically used for sending webhook notifications?

(a) GET

(b) POST

(c) DELETE

(d) PATCH

[Source ans: b]

2025 Aug31 ET

4. [Source Q2; MCQ]

In a message queue system, if the consumer is slower than the producer, what is likely to occur first?

(a) Messages will be dropped

(b) Messages will accumulate in the queue

(c) The consumer will speed up automatically

(d) The producer will block until the consumer is ready

[Source ans: b]

5. [Source Q8; MSQ]

Which of the following is/are the potential benefits of using a message broker?

(a) A message broker makes the network scalable for adding more servers to thenetwork.

(b) A message broker allows two servers in a network to directly communicatewith each other, without an intermediary.

(c) A message broker is not suited in case of traffic spikes, as messages areretained in the queue until processed.

(d) A message broker can be used for batch processing of messages.

[Source ans: a, d]

6. [Source Q9; MSQ]

Which statements about webhooks are correct?

(a) They enable real-time data delivery

(b) They use HTTP POST requests typically

(c) They're designed for server-to-server communication

(d) They require polling from the client

[Source ans: a, b, c]

7. [Source Q10; MCQ]

A popular e-commerce platform needs to notify multiple third-party services (inventory management, email marketing, analytics) whenever a customer places an order. The development team is considering different approaches to handle these notifications.

Scenario: When an order is placed, the system needs to:

• Update inventory levels in an external warehouse system • Send a welcome email through a third-party email service • Log analytics data to an external tracking service • Update customer loyalty points in a CRM system

Which of the following statements about using webhooks for this scenario is MOST accurate?

(a) Webhooks are not suitable for this use case because they require the e-commerce platform to continuously poll each third-party service to check if they're ready to receive data, which would create unnecessary network overhead.

(b) Webhooks should be avoided in this scenario because they operatesynchronously, meaning the customer's order placement would be delayed until all third-party services have successfully processed their notifications.

(c) Webhooks provide an ideal solution as they allow the e-commerce platform topush real-time notifications to all subscribed third-party services immediately when an order is placed, eliminating the need for these services to repeatedly poll for updates.

(d) Webhooks are primarily designed for read-only operations and cannot handlecomplex data payloads like order information, making them unsuitable for e-commerce transaction notifications.

[Source ans: c]

2025 Dec18 ET

8. [Source Q25; MSQ]

In a web application using webhooks and Redis for real-time data synchronization: Scenario: An order management system where: • When an order is created, a webhook is sent to a notification service • Redis stores the current order cache • Multiple concurrent requests update the cache Which statements about this architecture are true?

(a) Webhooks guarantee that the notification service will process events in the exact order they were triggered

(b) Redis is suitable for this use case because it's an in-memory data structure store that provides fast access to cached data

(c) The webhook will automatically retry if the notification service is temporarily unavailable

(d) Webhooks use HTTP POST requests to communicate events asynchronously

[Source ans: b, d]

2026 May06 ET

9. [Source Q4; MCQ]

Which of the following HTTP methods does WebSocket use for initial handshake?

(a) GET

(b) POST

(c) PUT

(d) DELETE

[Source ans: a]

2026 Sep13 ET

10. [Source Q3; MCQ]

A startup is building a food delivery app. The customer-facing screen shows live order status updates: "Order Placed → Preparing → Out for Delivery → Delivered". Each stage is triggered by the restaurant or the delivery rider updating the status on their end. The customer's screen should reflect these changes in real time without refreshing the page. Two developers debate about the right technology: Developer A says: "Let's use Server-Sent Events. It's the right fit here." Developer B says: "No, we must use WebSockets. SSE won't work." Who is correct, and why?

(a) Developer B is correct. SSE cannot transmit JSON data, so structured order status objects cannot be pushed to the browser.

(b) Developer B is correct. SSE connections do not reconnect automatically when the network is briefly interrupted, making it unreliable for a live tracker.

(c) Developer A is correct, but only if WebSockets are set up as a mandatory fallback, since SSE requires HTTP/2 to function.

(d) Developer A is correct. The tracker only requires the server to push status updates to the customer, without the customer sending data back through the same connection. SSE is purpose-built for exactly this one-way server-to-client push scenario.

[Source ans: d]

11. [Source Q7; MCQ]

A team builds an internal warehouse dashboard where multiple managers, each on a different computer, need to receive live low-stock alerts from the server in real time. They decide to use Flask-SSE with Redis. Which of the following best explains Redis's role in a Flask-SSE setup?

(a) Redis is used by Flask-SSE to authenticate connected users before events can be streamed to the browser.

(b) Redis is optional in Flask-SSE and is only recommended when the application has a very large number of concurrent users.

(c) Redis is only needed when SSE events must be stored permanently and replayed after reconnection.

(d) Flask-SSE uses Redis as a shared event channel so published messages can be delivered to connected clients.

[Source ans: d]